

Intractable Elevated Intraocular Pressure during Plaque Brachytherapy for Posterior Choroidal Tumors: A Case Report

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Keywords

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Abstract

Introduction: Uncontrolled elevated intraocular pressure (IOP) during brachytherapy for posterior choroidal tumors is unreported. We present 2 patients who both experienced intractable elevated IOP during plaque brachytherapy. **Case Presentation:** Both patients experienced persistent elevated IOP unresponsive to all pressure-lowering agents following surgery that included transvitreal retinochoroidal biopsy and ruthenium-106 plaque application for posterior choroidal tumors in the superior quadrants. During plaque placement in patient 1, a large, dilated vortex vein was observed in the area where the plaque was to be placed. After plaque removal in both patients, IOP normalized without the need for additional pressure-lowering therapy. **Conclusion:** We hypothesize that the clinical presentation in both patients was primarily attributable to two factors: first, pre-existing venous congestion and, second, the application of a plaque, which obstructed a superior vortex vein. This left the remaining vortex veins unable to compensate for the obstruction, leading to intractable IOP during plaque brachytherapy.

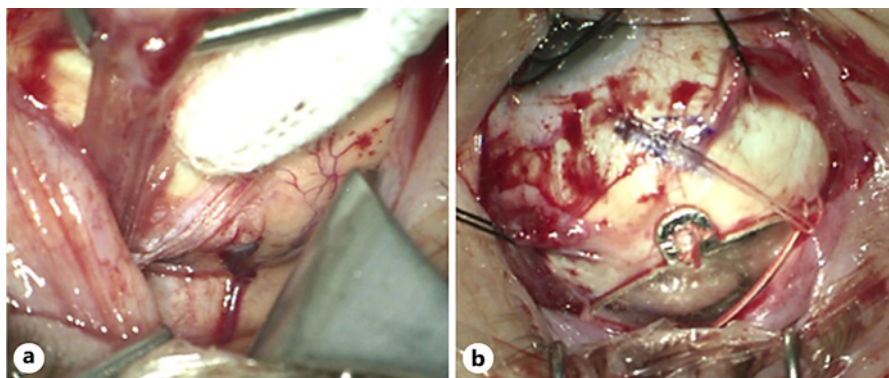
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Introduction

Plaque brachytherapy, as opposed to enucleation, is the preferred choice of treatment in small and medium-sized choroidal melanomas [1]. Brachytherapy-related complications can affect a wide range of ocular structures, with cataract, iris rubeosis, and glaucoma (open angle and neovascular types) being the most prevalent [2–5]. These complications are primarily long-term in nature. Among 73 patients having uveal melanoma, intraocular pressure (IOP) was found to increase on average 8.5 mm Hg during iodine-125 plaque brachytherapy compared to baseline, with the size of plaque being the only factor related to the increase in pressure [3]. We report 2 patients with a posterior choroidal tumor in which they both developed intractable elevated IOP during plaque brachytherapy. The IOP remained unresponsive to pressure-lowering treatment, e.g., systemic acetazolamide, oral glycerol, and topical eye drops.

In both patients, the IOP was first normalized after removal of the plaque. To the best of our knowledge, this complication to brachytherapy has not been reported before.

Fig. 1. a A large dilated superotemporal VV was present in the area for plaque placement in patient 1. **b** After plaque placement.



Case Presentation

Patient 1

A 77-year-old female with a superotemporal choroidal tumor in her right eye was scheduled for a minimally invasive transvitreal retinochoroidal biopsy using a 25-gauge vitrector system and placement of a 20.2 mm ruthenium-106 plaque (CCB) in June 2024 [6]. The tumor height measured by ultrasound was 4.3 mm. Ocular history included amblyopia in her right eye (best-corrected visual acuity was 20/400) and hyperopia (+1.5 diopters/axial length, 22.4 mm in both eyes). Medical history included breast cancer 3 years prior treated with surgery and radiotherapy and Wolff-Parkinson-White syndrome.

Perioperatively before plaque placement, a large and dilated vortex vein (VV) was identified, as shown in Figure 1. The patient was seen for routine follow-up the day after surgery. The applanatory IOP was 54 mm Hg and was lowered to 28 mm Hg with intravenous acetazolamide, oral glycerol (1.5 mg/kg body weight), and pressure-lowering eye drops (timolol, dorzolamide, latanoprost, brimonidine, and netarsudil). By mistake, topical pilocarpine was also administered and an additional increase in pressure was observed. The IOP stayed resistant to pressure-lowering treatment and continued to stay increased (34–60 mm Hg) for the next 3 days. On the fifth postoperative day, the plaque was removed according to the radiation treatment plan, and the pressure normalized (10 mm Hg) without the need for additional pressure-lowering agents. The intraoperative biopsy verified the diagnosis of choroidal melanoma. Six months following plaque brachytherapy, tumor height decreased to 1.1 mm, but the patient had no light perception, and the optic disc appeared pale and excavated. The patient remained without light perception at follow-up after 16 months (Fig. 2).

Patient 2

An 81-year-old female with a choroidal tumor located superior to the optic disc in her right eye was also scheduled for a minimally invasive transvitreal retinochoroidal biopsy using a 25-gauge vitrector system and a 19.8 mm notched ruthenium-106 plaque (COB) in July 2024. No dilated VVs were noted during surgery. The tumor height was 1.9 mm. Ocular history was unremarkable, and she was emmetropic with a normal axial length. Medical history included disseminated breast cancer for 4.5 years, type 2 diabetes, hypertension, hypercholesterolemia, and collagen colitis.

Postsurgical ophthalmic examination on the same day revealed an IOP of 60 mm Hg. Like patient 1, IOP could not be controlled (>36 mm Hg) despite use of systemic acetazolamide, oral glycerol and pressure-lowering eye drops (timolol, dorzolamide, latanoprost, brimonidine and netarsudil). The plaque was removed on postoperative day 2 according to the radiation treatment plan, which normalized the pressure (18 mm Hg), and no additional pressure-lowering agents were needed. The intraoperative biopsy verified a metastasis from her breast cancer. Six months after plaque brachytherapy, the tumor height had shrunk to 0.7 mm, and the best-corrected visual acuity was 20/20 (Fig. 2).

Discussion

We report 2 patients who exhibited a similar clinical course of elevated IOP, refractory to all available pressure-lowering agents, during plaque brachytherapy for a posterior choroidal tumor. In patient 1, a significantly dilated superior VV was noted before placement of the plaques. Notably, IOP normalized following plaque removal in both patients. Although no dilated VVs were observed during surgery in patient 2, we hypothesize that a similar mechanism may underline both cases as no comparable clinical presentation has been observed at our institution or reported elsewhere.

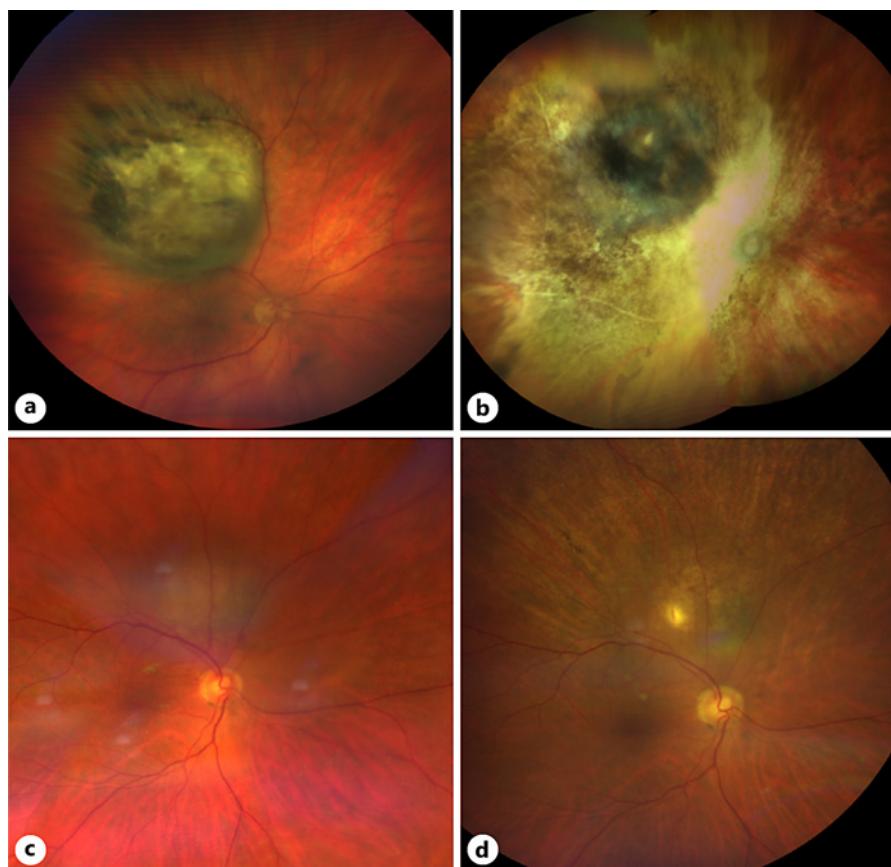


Fig. 2. Clarus 500™ ultra-widefield fundus images obtained before and after surgery. Patient 1 prior to plaque placement (**a**) and 16 months post-plaque placement (**b**). A choroidal melanoma is seen superotemporally. Patient 2 before plaque placement (**c**) and 6 months post-plaque placement (**d**). A choroidal metastasis, originating from disseminated breast cancer, is visible superior to the optic disc.

The observation of a single compromised superior VV causing such a clinical outcome was unexpected. In 1973, the consequence of experimental VV occlusion was studied in rhesus monkeys. An immediate increase in IOP of 40–60 mm Hg was noted if all VV were occluded; however, occlusion of one or two VV did not produce a similar effect [7]. IOP normalized after 1 week in eyes with complete VV occlusion. Interestingly, VV anastomoses among the horizontal watershed zone are a common feature among normal eyes [8] and were observed preoperatively in patient 2. These anastomoses should theoretically have mitigated the venous overload. A study utilizing widefield optical coherence tomography angiography demonstrated that the VV drainage preferentially occurs in the following order across the four quadrants: superotemporal, superonasal, inferotemporal, and inferonasal [9]. In our patients, a superior VV (or possibly both superior VV in patient 2) was obstructed by the ruthenium-106 plaque. Furthermore, a study has shown that short axial length is associated with congested VV in which a thickened sclera was hypothesized to disturb venous outflow among patients with central serous chorioretinopathy [10]. In line with this hypothesis, the shorter axial length of patient 1 may have

contributed to the increase in IOP. We, therefore, hypothesize that the clinical presentation observed in both patients resulted from a sequence of events: (1) pre-existing venous congestion (evidenced by dilated VV before plaque placement in patient 1), (2) obstruction of a superior VV by the plaque, (3) inadequate compensatory drainage by the remaining VV, and (4) a potential contribution by a thickened sclera, particularly in patient 1, exacerbating the venous congestion. The short duration of the obstruction may have impeded compensatory mechanisms to form, which may require at least 1 week, as noted in the rhesus monkey study.

These two cases highlight that plaque brachytherapy for posterior choroidal tumors may result in intractably elevated IOP. In such cases, the removal of the plaque appears to be the only effective intervention for normalizing IOP in the acute phase.

Statement of Ethics

Ethical approval was not required for this case report. Upon consultation, the Regional Ethics Committee (Capital Region of Denmark) confirmed that formal approval was

unnecessary. Written informed consent was obtained from both patients for publication of their medical case and accompanying images.

Conflict of Interest Statement

No authors have any conflict of interest to declare.

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Author Contributions

Concept, design, and drafting of the manuscript: Mohamed Belmouhand and Carsten Faber. Acquisition, analysis or interpretation of data and final approval of the manuscript: Mohamed Belmouhand, Jens Folke Kiilgaard, and Carsten Faber. Critical revision of the manuscript for important intellectual content: Jens Folke Kiilgaard.

Data Availability Statement

The data supporting the findings of this study are not publicly available due to data protection regulations but can be obtained from the corresponding author upon request.